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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$. Find m

Select the correct option

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- ○ 3
- ◉ 4 (Module 132, Pg.57)
- ○
- ○

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$. Find m

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

Velocity of A before collision $= u_1 = 4\hat{i}$

Velocity of B before collision $= u_2 = 3\hat{i}$

Velocity of A after collision $= v_1 = 5\hat{i}$

Velocity of B after collision $= v_2 = 2\hat{i}$

Mass of Particle $A = m_1 = 4$

Mass of Particle $B = m_2 = ?$

Using the principle of conservation of momentum:

$$m_1(u_1 - v_1) = m_2(v_2 - u_2)$$

$$4(4 - 5) = m_2(2 - 3)$$

$$-4 = -m_2$$

$$4 = m_2$$

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If a body rotates about some external fixed point it is called revolution

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Select the correct option

- spin
- revolu
-
-

If the axis of rotation passes through the center of mass of the rigid body then body is said to be spin or rotate upon itself. If a body rotates about some external fixed point is called revolution or orbital motion of the rigid body.

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If a body rotates about some external fixed point it is called revolution

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Select the correct option

- spin
- revolution

(Module 122, Pg.42)

-
-

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

Angular momentum can be defined as

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Select the correct option

- $L =$
- $L =$
- $L =$
- $L =$

Angular Momentum

The angular momentum of a single particle is defined as the cross product of linear momentum and position vector of concerned particle.

Mathematically

$$L = r \times p$$

$$L = r \times mv$$

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

Angular momentum can be defined as

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Select the correct option

- $L = r \times v$
- $L = r \times \omega$
- $L = p \times \omega$
- $L = r \times p$

(Module 135, Pg.64)

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

The moment of inertia of a particle about y-axis is

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Select the correct option

- $I_{yy} = m(x^2 + \quad ^2)$
- $I_{yy} = mx$
-
-

The moment of inertia of a particle of mass m with coordinates (x, y, z) relative to the orthogonal Cartesian coordinate system $OXYZ$ about X, Y, Z axes will be

X-axis: $I_{xx} = m(y_2 + z_2)$

Y-axis: $I_{yy} = m(z_2 + x_2)$

Z-axis: $I_{zz} = m(x_2 + y_2)$

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

The moment of inertia of a particle about y-axis is

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Select the correct option

- $I_{yy} = m(x^2 + y^2)$ (Module 125, Pg.46)
- $I_{yy} = mxz$
-
-

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

The torque acting on a particle is proportional to the time rate of change in its angular acceleration.

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Select the correct option

- false
- true

Theorem:

The torque acting on a particle equals the time rate of change in its angular momentum, i.e.,

$$\tau = \frac{d\Omega}{dt}$$

where $\Omega = r \times p$ is defined angular momentum of the system.

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

The torque acting on a particle is proportional to the time rate of change in its angular acceleration.

Select the correct option

PRESENTED BY: TALEEMI MARKAZ

- false (Module 119, Pg. 38)
- true
-
-

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$.

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Select the correct option

- ○ $I =$
- ○ $I =$
- ○
- ○

As we know that:

$$K^2 = \frac{I}{M} \quad \ldots Eq.(1)$$

Given that:

$$K = \sqrt{\frac{2}{5}a}$$

By putting the value of k in Eq.(1), we get

$$\left(\sqrt{\frac{2}{5}a}\right)^2 = \frac{I}{M} \quad \Rightarrow \frac{2}{5}a^2 = \frac{I}{M}$$

$$\frac{2}{5}Ma^2 = I$$

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$.

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Select the correct option

- ○ $I = \frac{5M}{2a^2}$
- ● $I = \frac{2}{5}Ma^2$ (Module 127, Pg.49)
- ○
- ○

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow _____

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Select the correct option

- a pa
- a str
-
-

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow a parabola.

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow _____

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Select the correct option

- a parabola

(Module 124, Pg.45)

- a straight line
-
-

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

The product of inertia of a particle about x and y-axis is

Select the correct option

PRESENTED BY: TALEEMI MARKAZ

- $I_{xy} =$
- $I_{xy} =$
-
-

The product of inertia for the same particle w.r.to the pair of coordinate axes are defined as

X and Y axis: $I_{xy} = -mxy$

Y and Z axis: $I_{yz} = -myz$

Z and X axis: $I_{zx} = -mzx$

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

The product of inertia of a particle about x and y-axis is

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- $I_{xy} = -mxy$ (Module 125, Pg.46)
- $I_{xy} = m(x^2 + y^2)$
-
-

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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

The position vector of the center of mass of a system of particles is

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Select the correct option

- ○ $\vec{r}_c =$
- ○ $\vec{r}_c =$
- ○
- ○

If m_i is the mass and v_i the velocity of the ith particle, then the kinetic energy of the ith particle will be $T_i = \frac{1}{2} m_i {v_i}^2$

Now by using the definition of position vector center of mass (c.m.), we have

$$\vec{r}_c = \frac{\Sigma_i m_i \vec{r}_i}{\Sigma_i m_i}$$

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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

The position vector of the center of mass of a system of particles is

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- $\vec{r}_c = \frac{\Sigma_i m_i \vec{r}_i}{\Sigma_i \vec{r}_i}$
- $\vec{r}_c = \frac{\Sigma_i m_i \vec{r}_i}{\Sigma_i m_i}$ (Module 141, Pg.76)
-
-

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Question # 2 of 10

SOLVED BY: WAQAR HUSSAIN

The radius of gyration K of the hemisphere of mass M and moment of inertia $I = \frac{2}{5}Ma^2$ about an axis through its center is

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Select the correct option

- ○ $K = \sqrt{$
- ○ $K = \frac{1}{a}$
- ○
- ○

As we know that:

$$K^2 = {}^{I}/_{M}$$

$$K^2M = I \quad \ldots Eq.(1)$$

Given that: $I = \frac{2}{5}Ma^2$

By putting the value of I in Eq.(1), we get

$$K^2M = \frac{2}{5}Ma^2$$

$$\sqrt{K^2} = \sqrt{\frac{2}{5}a^2}$$

$$K = \sqrt{\frac{2}{5}}a$$

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Question # 2 of 10

SOLVED BY: WAQAR HUSSAIN

The radius of gyration K of the hemisphere of mass M and moment of inertia $I = \frac{2}{5}Ma^2$ about an axis through its center is

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Select the correct option

- $K = \sqrt{\frac{2}{5}}a$ (Module 127, Pg.49)
- $K = \frac{1}{a}\sqrt{\frac{5}{2}}$
-
-

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

In general, the directions of angular velocity and angular momentum are

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Select the correct option

- same
- dist
-
-

When a rigid body is rotating about a fixed point O, the angular velocity vector ω and the angular momentum vector L (about O) are not in general in the same direction.

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

In general, the directions of angular velocity and angular momentum are

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Select the correct option

- same
- distinct

(Module 128, Pg.50)

-
-

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass 5 with velocity $3\hat{i}$. Then what will be the velocity of first particle after collision if the velocity of second particle after collision is $2\hat{i}$.

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Select the correct option

Velocity of A before collision $= u_1 = 4\hat{i}$

Velocity of B before collision $= u_2 = 3\hat{i}$

Velocity of A after collision $= v_1 = ?$

Velocity of B after collision $= v_2 = 2\hat{i}$

Mass of Particle $A = m_1 = 4$

Mass of Particle $B = m_2 = 5$

Using the principle of conservation of momentum:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$4(4) + 5(3) = 4v_1 + 5(2)$$

$$31 - 10 = 4v_1$$

$$v_1 = {}^{21}/_{4}\,\hat{i}$$

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass 5 with velocity $3\hat{i}$. Then what will be the velocity of first particle after collision if the velocity of second particle after collision is $2\hat{i}$.

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Select the correct option

- $\frac{21}{4}\hat{i}$ (Module 132, Pg.57)
- $\frac{15}{4}\hat{i}$
-
-

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

For an elastic body, the distance between two particles _____ under the action of applied force

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Select the correct option

- ○ remai
- ○ is cha
- ○
- ○

When a force is applied to a system of particles, it changes the distance between individual particles. Such systems are often called deformable or elastic bodies.

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

For an elastic body, the distance between two particles _____ under the action of applied force

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Select the correct option

- remains constant
- is changed

(Module 121, Pg.41)

-
-

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

For inertia matrix, along principal axes

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Select the correct option

- the off diagonal elements are zero
- th
-
-

A matrix with all zero off-diagonal elements is, as we said, called a diagonal matrix. We can then write

$$L = I\omega = (\lambda_1\omega_x, \lambda_2\omega_y, \lambda_3\omega_z)$$

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

For inertia matrix, along principal axes

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Select the correct option

- the off diagonal elements are zero (Conceptual)
- the diagonal elements are zero
-
-

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

If angular momentum $\Omega = 2\omega \sin \omega t$ then the torque τ is

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Select the correct option

- -2
- 2ω
-
-

As we know that:

$$\vec{\tau} = \frac{d\vec{\omega}}{dt}$$

Given that: $\Omega = 2\omega \sin \omega t$

By taking derivative, we get

$$\vec{\tau} = \frac{d}{dt}(2\omega \sin \omega t)$$

$$\vec{\tau} = 2\omega \cos \omega t \ \frac{d}{dt}(\omega t)$$

$$\vec{\tau} = 2\omega \cos \omega t \ \frac{d}{dt}(\omega t)$$

$$\vec{\tau} = 2\omega \cos \omega t \ (\omega)$$

$$\vec{\tau} = 2\omega^2 \cos \omega t$$

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

If angular momentum $\Omega = 2\omega \sin \omega t$ then the torque τ is

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Select the correct option

- $-2 \cos \omega t$
- $2w^2 \cos \omega t$ (Conceptual)
-
-

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- ○ $I =$
- ○ $I =$
- ○
- ○

As we know that:

$$K^2 = \frac{I}{M} \quad \ldots Eq.(1)$$

Given that:

$$K = \sqrt{\frac{2}{5}}a$$

By putting the value of k in Eq.(1), we get

$$\left(\sqrt{\frac{2}{5}}a\right)^2 = \frac{I}{M} \quad \Rightarrow \frac{2}{5}a^2 = \frac{I}{M}$$

$$\frac{2}{5}Ma^2 = I$$

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$.

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Select the correct option

- ○ $I = \frac{5M}{2a^2}$
- ● $I = \frac{2}{5}Ma^2$ (Module 127, Pg.49)
- ○
- ○

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

Dynamics deals with the forces acting on any object and relationship fundamentally to the motion.

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Select the correct option

- true
- false

Dynamics is the branch of mechanics deals with forces and relationship fundamentally to the motion but sometimes also to the equilibrium of bodies.

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

Dynamics deals with the forces acting on any object and relationship fundamentally to the motion.

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Select the correct option

- true (Module 129, Pg.51)
- false
-
-

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

The product of inertia of a particle about x and y-axis is

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Select the correct option

- ○ $I_{xy} =$
- ○ $I_{xy} =$
- ○
- ○

The product of inertia for the same particle w.r.to the pair of coordinate axes are defined as

X and Y axis: $I_{xy} = -mxy$

Y and Z axis: $I_{yz} = -myz$

Z and X axis: $I_{zx} = -mzx$

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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

Find the time period T of the vibrating mass with $m = 5$ and $f = 10$

Select the correct option

It is measure in hertz (Hz).

The frequency of S.H.M can be calculate by using the following relation

$$f = \frac{1}{T}$$





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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

Find the time period T of the vibrating mass with $m = 5$ and $f = 10$

Select the correct option

- $T = \frac{1}{10}$ (Module No.103 Pg.13)
- $T = \frac{1}{2}$
-
-

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Question # 2 of 10

SOLVED BY: WAQAR HUSSAIN

If position of a vibrating mass at any time is defined as $y(t) = 10sin\left(\frac{1}{3}t\right)$. Then the amplitude is

Select the correct option

iv. Amplitude, T, F ?

General form for the position of a particle moving to and fro is

$$y(t) = \mathrm{A}\cos(2\pi t/\mathrm{T})$$

Thus, $y = 10\cos\sqrt{3/2}t$

Amplitude $= 10$

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Question # 2 of 10

SOLVED BY: WAQAR HUSSAIN

If position of a vibrating mass at any time is defined as $y(t) = 10sin\left(\frac{1}{3}t\right)$. Then the amplitude is

Select the correct option

- $\frac{1}{3}$
- 10

(Module No.104 Pg.15)

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

Gravitational force is an example of a conservative force.

Select the correct option

- t

Examples of Conservative Forces

- Gravitational force is an example of a conservative force.
- Elastic spring force is example of conservative force.

- f
-
-

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

Gravitational force is an example of a conservative force.

Select the correct option

- true (Module No.97 Pg.35)
- false
-
-

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

The trajectory of a particle in projectile motion with air resistance is

Select the correct option

- a parabola

Consider now a spherical object launched with a velocity V forming an angle theta with the horizontal ground. In the absence of air resistance, the trajectory followed by this projectile is known to be a parabola. This follows from writing Newton's law separately for the horizontal and vertical coordinates. The former scales linearly with time whereas the latter varies quadratically.

- none of these

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

The trajectory of a particle in projectile motion with air resistance is

Select the correct option

- a parabola
- a non parabolic curve
- a hyperbola
- none of these

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

The maximum distance covered by the oscillating body in one oscillation or length of a wave measured from its mean position is called amplitude.

Select the correct option

Amplitude

Amplitude is defined as the maximum distance covered by the oscillating body in one oscillation or length of a wave measured from its mean position.

The amplitude of a pendulum is one-half the distance that the mass covered in moving from one terminal to the other. The vibrating sources generate waves, whose amplitude is proportional to the amplitude of the vibrating source.

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

The maximum distance covered by the oscillating body in one oscillation or length of a wave measured from its mean position is called amplitude.

Select the correct option

- false
- true

(Module No.103 Pg.13)

2:15 / 4:40

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- Forces that cannot be expressed in the term of a potential energy function are called non-conservative forces. We can also state that forces that do not store energy are called non-conservative or dissipative forces. If there is no scalar function V such that $F = -\Delta V$ [or, equivalently, if $V \times F = 0$], then F is called a non-conservative force field.
-
-
-

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

Forces that can be expressed in the term of a potential energy function are called non conservative forces.

Select the correct option

- true
- false

(Module No.100 Pg.39)

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

The force field $\vec{F} = 3x\hat{i} - 2y\hat{j}$ is

Select the correct option

- nc
- cc
- As we obtain $\nabla \times \vec{F} = 0$
- We conclude that $\vec{F}$ is conservative.
-
-

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

The force field $\vec{F} = 3x\hat{i} - 2y\hat{j}$ is

Select the correct option

- non conservative
- conservative

(Module No.98 Pg.36)

-
-

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

A car driving along a curved road is an example of

Select the correct option

Example

- A stone thrown into the air at an angle.
- A car driving along a curved road.
- Throwing paper airplanes or paper darts is an example of curvilinear motion.

- curvilinear motion
- none of these options

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

A car driving along a curved road is an example of

Select the correct option

- rectilinear motion
- oscillatory motion
- curvilinear motion

(Module No.112, Pg.27)

- none of these options

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

Friction is a

Select the correct option

Friction is a non-conservative force, and there are others. It is always opposed to the direction of motion and is not a single valued function of position alone.

Similarly the impulse (time dependent force) is also non-conservative and cannot be derived from a scalar point function.

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

Friction is a

Select the correct option

- non conservative force (Module No.100, Pg.39)
- conservative force
-
-

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line followed by a _____ about that line.

Select the correct option

Theorem Statement:

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line (called its screw axis) followed (or preceded) by a rotation about that line.

- ○ [illegible]
- ○ rotation
- ○ displacement

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line followed by a _____ about that line.

Select the correct option

- translation
- reflection
- rotation (Module No.109, Pg.23)
- displacement

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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

Dynamics deals with the forces acting on any object and relationship fundamentally to the motion.

Select the correct option

- false
- true (Module No.129 Pg. 51)
-
-

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Question # 2 of 10

SOLVED BY: WAQAR HUSSAIN

The moment of inertia of a particle about y-axis is

Select the correct option

- $I_{yy} = m(x^2 + y^2)$ (Module No.125 Pg.46)
- $I_{yy} = mxz$
-
-

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

Consider a right circular cone of density ρ, radius α and height h is composed of elementary circular discs of small thickness each parallel to the base of the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz then mass of the disc is

Select the correct option

- $\delta m = \rho \pi r^2 \delta z$ (Module No.139 Pg.72)
- $\delta m = \rho 2\pi \delta z$
-
-

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If the axis of rotation passes through the center of mass of the rigid body then it is called

Select the correct option

- revolution
- spin

(Module No.128 Pg.50)

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

The product of inertia of a particle about x and y-axis is

Select the correct option

- $I_{xy} = m(x^2 + y^2)$
- $I_{xy} = -mxy$ (Module No.125 Pg.46)
-
-

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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

If an object is thrown into the air, different parts of the object can follow quite complicated paths, but the center-of-mass will follow _____

Select the correct option

- a parabola (Module No.124 Pg.45)
- a straight line
-
-

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

Kinetic energy T of a rigid body in a planar motion, is defined by

Select the correct option

- $T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$ (Module No.137 Pg.68)
- $T = \frac{1}{2}I_o\omega^2$
-
-

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

If a rigid body is rotating about a fixed point on ground then its kinetic energy T is defined by

Select the correct option

- $T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2$
- $T = \frac{1}{2}I_o\omega^2$ (Module No.137 Pg.68)
-
-

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Question # 9 of 10

SOLVED BY: WAQAR HUSSAIN

Angular momentum can be defined as

Select the correct option

- $L = r \times p$ (Module No.135 Pg.64)
- $L = r \times v$
- $L = r \times \omega$
- $L = p \times \omega$

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Question # 10 of 10

SOLVED BY: WAQAR HUSSAIN

If P is any point on the ellipsoid of inertia, then the moment of inertia of a rigid body about any line $\overrightarrow{OP}$ is equals to the reciprocal of the square of the length of $\left|\overrightarrow{OP}\right|$.

Select the correct option

- false
- true

(Module No.140 Pg.75)

-
-

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n # 3 of 5 (Start time: 11:08:22 AM, 03 March 2022) Total Mar

Moment of inertia of a hollow sphere having mass M and radius a is given by

he correct option

Reload Math Equat

$I = \frac{1}{2}Ma^2$

$I = Ma^2$

$I = \frac{2}{5}Ma^2$

$I = \frac{2}{3}Ma^2$

tion # 2 of 5 (Start time: 11:07:32 AM, 03 March 2022) Total M

he center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

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ct the correct option Reload Math Equa

coincident



perpendicular

Click to Save Answer & Move to Next Question

MTH622:QUIZ NO.3

Quiz Start Time: 11:06 AM, 0

Question # 4 of 5 (Start time: 11:09:44 AM, 03 March 2022)

Consider a right circular cone of density ρ, radius a and height h is composed of elementary circular discs of small thickness each parallel to the the cone. Choose the z-axis as the axis of symmetry and consider a typical disc of radius r and width δz them mass of the disc is

Select the correct option

Reload M

- $\delta m = \rho 2\pi r \delta z$
- $\delta m = \rho \pi r^2 \delta z$

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Question # 5 of 10 (Start time: 03:02:20 PM, 23 February 2021)

The Moment of inertia of a solid sphere having mass M and radius a is given by

Select the correct option

O	$I = Ma^2$
O	$I = \frac{1}{2}Ma^2$
O	$I = \frac{2}{5}Ma^2$

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Module 136

12:10 PM